



FNB DataQuest Toolkit

From Roots to Rise



INTRODUCTION

Welcome to the FNB DataQuest!

We're excited to host you in this 1-day event which will allow you to learn, network and most importantly put your mind to the test against the best and the brightest in the data science community.

Better data means better analytics and better ways to help our customers. With better insights come better products and services, which makes data and analytics critical to what we do as a business and why we need the sharpest minds to do it. Through quantitative, analytical, mathematical, and statistical methods, we are better able to make sense of data, understand business trends, make smart decisions, and solve financial and risk problems, credit modelling and pricing.

As part of the FNB DataQuest, we've given you real challenges that our own analytics teams have had to solve, so that you can get an idea of life at FNB as an analyst. They're not easy, but to help guide you through the process we have also provided some very useful masterclasses and the opportunity to ask our subject matter experts questions. You will also have a chance to connect with our experts in 'coaching' rooms.

At FNB, we're always looking for individuals who've got the energy, enthusiasm, and experience to make a difference. We're excited to spend time with you and give you the opportunity to show us what you've got!

Please read through the rest of the toolkit as there are a few things you need to know before you get ready to solve, learn and win.

The FNB Young Talent Team

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AGENDA

5 May

- 09:00 Welcome
- 09:15 Keynote address
- 10:30 Challenge Introduction,
Toolkit Review and Challenge Q&A
- 11:30 Born to be Great Game Show
- 13:00 Lunch
- 13:30 Masterclass and Challenge Q&A
- 14:30 Closing

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CHALLENGE

Interpretable Credit Modelling

Context

You are part of the analytics team at a retail lending company that provides consumer loans. The company wants to improve how it evaluates loan applications using a historical dataset of past applicants, their application-time information, and whether they defaulted within 12 months.

The business requires an interpretable solution — specifically logistic regression — that can be explained to risk managers, stakeholders, and regulators. This mirrors real-world credit modelling constraints where transparency and regulatory compliance are paramount.

Problem Statement

Build an interpretable predictive model that helps the lender make better approval decisions, while also providing interactive tools for data exploration and business decision support. The model must be logistic regression, improved through thoughtful feature engineering informed by exploratory analysis.

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Questions

1 Research & Interactive EDA Tool

Research credit modelling concepts (GLMs vs non-linear models, WoE/IV, key metrics, regulatory considerations) and build an interactive app (e.g. Streamlit/Shiny) with a univariate explorer (distributions, WoE, IV), a bivariate explorer (scatterplots, heatmaps), and a data quality report.

2 Improve Baseline Logistic Regression

Using EDA findings, apply feature engineering and scorecard-style transformations to push logistic regression performance beyond the baseline AUC of 0.68, benchmarked against a LightGBM ceiling of 0.82. Clearly explain what was discovered, what was adjusted, and what improvement was realised.

3 Business Value Dashboard (Bonus)

Extend the app to show how the model supports lending decisions: volume vs risk trade-offs at different thresholds, precision/ recall in business terms, and policy-level thinking.

Deliverables

- ✓ Research & Interactive app code (EDA + model evaluation + dashboard)
- ✓ Modelling summary report (research, EDA findings, cleaning, feature engineering, methodology, final model equation, performance, business recommendations)
- ✓ Reproducible workflow with attached code
- ✓ AI usage reflection (prompts used, what helped, what was accepted/modified/rejected)

Constraints

- ! Final model must be logistic regression
- ! Emphasis on interpretability and clear reasoning
- ! EDA must guide modelling choices
- ! No automated feature selection without explanation

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COMPETITION

Rules

- 1 Participants can work individually or in teams, one person can choose to represent the team in the submission presentation.
- 2 Submission must be in PowerPoint format, with attached script and an embedded video presentation.
- 3 Any person who is at least 18 years old with a valid student number and meets the following criteria may take part:
 - must be a South African citizen;
 - must be a full-time student currently in their final year of studying towards a degree or a postgraduate degree;
 - must at the time of the competition be in their second, final or postgraduate degree to study at any South African university.

How to submit

Entries need to be submitted before **22 May 2026** at 23:59. Submissions will need to consist of an Explanation Video/ PowerPoint and support files. We suggest uploading these to your Google Drive, Microsoft OneDrive or Dropbox and sharing the link on the submission form below:

Click here to submit or copy and paste the link in your browser: <https://tinyurl.com/dataquestSubmit>.

Please ensure you have reviewed the competition rules above and accept the *full terms and conditions*, **click here** or copy and paste the link in your browser the link: <https://tinyurl.com/datatermsandc>.

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COMPETITION

Judging Weighting

Category	Weighting %	Rationale
Research & Conceptual Understanding	25%	The project is fundamentally about learning credit modelling concepts and demonstrating understanding
EDA Tool & Data Investigation	30%	Discovery is the core skill - the planted effects are only valuable if found through genuine exploration
Feature Engineering & Modelling	20%	The model itself is constrained to logistic regression; what matters is the reasoning behind transformations
Business Value & Communication	15%	Translating model output into decisions is the real-world payoff
AI Usage & Process	10%	Honest reflection on AI as a tool, not as substitute

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Prizes

1st Prize

The first prize winner will receive a **R25 000 cash prize**, as well as a **guaranteed interview** for the FNB Future League Graduate Programme

2nd Prize

Second prize winner will receive **R10 000**.

3rd Prize

The third prize winner will receive **R5 000**.

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Resources

The DataQuest event will be hosted on the **Microsoft Teams** platform.

DATA FILES REQUIRED FOR THE CHALLENGE:

Challenge: Interpretable Credit Modelling

Click here for data files or copy and paste the link in your browser: <https://tinyurl.com/dqChallenge1>

DataQuest Day Recording

Click here or copy and paste the link in your browser: <https://tinyurl.com/DataQuest2026Day1>

Contacts

If you have any questions, email grad@fnb.co.za.

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